PLANRS ACADEMY LESSON PLAN

■ Shadow Science ■

Discover the Science of Light & Shadows

Grade: Grade 2 (Age 7–8)	**Subject:** STEM — Shadow Science	**Duration:** 30 Minutes
Level: Beginner	**Session:** Session 4 of 5	**Format:** Interactive + Story + Activity

■ LEARNING OBJECTIVES

LO1	Explain that shadows form when an object blocks light.
LO2	Identify opaque, translucent, and transparent materials and predict shadow darkness.
LO3	Observe how shadow size changes based on the distance of the light source.
LO4	Connect shadow science to Indian cultural examples (shadow puppetry, sundials).
LO5	Complete an experiment and record observations like a science detective.

■■ SESSION TIMELINE (30 Minutes)

Time	Phase	Description
00:00–04:00	WARM-UP	Detective Badge Award + Quick Review Quiz (3 questions from Session 3)
04:00–10:00	CONCEPT	Opaque / Translucent / Transparent — Animated explainer with Indian examples
10:00–22:00	ACTIVITY	Shadow Science Detective Lab — Interactive HTML experiment activity
22:00–27:00	HANDS-ON	Shadow Puppet Mini Project — Cut & create a puppet with black card
27:00–30:00	WRAP-UP	Gallery share + Exit ticket: "Name one opaque thing in your home"

■ KEY CONCEPTS

■ Opaque Materials

- Block ALL light completely
- Cast dark, solid shadows
- Examples: tiffin box, thick book, wooden board, black card

■ Translucent Materials

- Let SOME light pass through
- Cast faint, blurry shadows
- Examples: butter paper, frosted glass, thin dupatta fabric

■ Transparent Materials

- Let MOST light pass through
- Cast almost no shadow
- Examples: clear glass, clean water in glass, plastic wrap

■ Shadow Size & Distance

- Light CLOSE to object → BIG shadow
- Light FAR from object → SMALL shadow
- Noon sun above → shortest shadow of the day

■■ INDIAN CONTEXT & EXAMPLES

■ **Tholpavakoothu**
Kerala shadow puppet theatre uses thick leather puppets (opaque) behind a lit cloth screen (translucent). The lamp behind creates sharp moving shadows — science in art!

■ **Jantar Mantar Sundial**
Built in Jaipur centuries ago, the world's largest sundial uses shadow direction and length to tell the time — proof that shadow science has Indian roots!

■ **Kite Festival Shadows**
During Makar Sankranti, kites flying low cast bigger shadows on the ground than kites flying high — a perfect real-life demo of shadow size and distance!

■ **Jhali Windows**
Traditional Indian lattice windows (jhali) are semi-opaque — they create beautiful patterned shadows on the floor, combining art and translucency.

■ HANDS-ON EXPERIMENT — Shadow Puppet Creation

What you need	Black thick card/paper (or dark coloured cardboard), scissors, a stick or ruler, a torch or phone flashlight, a plain white wall or paper screen.
Cost Estimate	Under ■20 (black card is widely available at stationery shops). Alternatives: cut the back of a used notebook.
Step 1	Cut a simple shape from the black card — a hand, an animal silhouette, a leaf, or any flat shape. Attach it to the stick with tape.
Step 2	Darken the room. Hold the torch steady and hold your puppet between the torch and the wall/screen.
Step 3 — Experiment!	MOVE THE PUPPET CLOSE to the torch. What happens to the shadow? Now move it FAR from the torch. What changes?
Step 4 — Predict & Record	Before moving, PREDICT what will happen (big/small, sharp/blurry). Then test it. Draw the shadow sizes in your journal.
Science Explanation	When the puppet is close to the light, it blocks a wider ANGLE of light rays → BIG shadow. Far away → narrower angle → SMALL shadow.
Cultural Link	You are now a shadow puppeteer! This is the same technique used in Kerala's Tholpavakoothu puppet shows for hundreds of years.

■ SESSION ASSESSMENT — Exit Ticket

#	Assessment Question	Answer Space
Q1	A steel tiffin box, a glass of water, and butter paper are placed in front of a torch. Which one makes the DARKEST shadow? Explain why.	______________________
Q2	Draw what happens to the shadow of a pencil when you move the torch CLOSER to it. Use arrows to show the change.	______________________
Q3	True or False: A transparent object makes the same shadow as an opaque object. Explain your answer.	______________________
Q4	You want to make a shadow puppet show. Should you make the puppet from glass or from thick black card? Give a reason.	______________________
Q5	At noon, a stick makes a short shadow right below it. At evening, the shadow becomes long. Why does this happen?	______________________

■■■ TEACHER / FACILITATOR NOTES

■■ Common Misconceptions

- Misconception: "Shadow is the reflection of an object." Correction: A shadow is the ABSENCE of light — it is a dark area, not a reflection.
- Misconception: "Only the Sun makes shadows." Correction: Any light source (torch, diya, tube light) makes shadows.
- Misconception: "Shadow always looks exactly like the object." Correction: Shadow shape depends on the ANGLE of light — it can stretch or compress.
- Misconception: "Transparent things have no shadow at all." Correction: They have a very faint shadow — light is slightly bent, not completely passed through.

■ Differentiation Tips

- Faster learners: Ask them to design a shadow clock using two sticks and predict what time it shows based on the shadow.
- Slower learners: Provide a printed "Shadow Sort" sheet with pictures to categorise before the digital activity.
- ELL students: Use the regional language vocabulary panel — shadow in their home language connects the concept.
- Apartment students: Use a phone torch in a dark bathroom — works perfectly as a substitute for sunlight experiments.
- Extension: Can students create a shadow that is LARGER than their whole arm? Challenge them to find the right distance!

■ **Low-Cost / No-Cost Alternatives:** No torch? Use a phone screen set to flashlight mode. No black card? Use the back of a dark notebook cover. No screen? Use a plain wall in a darkened room. No clay for a stick? Use Play-Doh or a lump of roti dough!

■ VOCABULARY GLOSSARY

English Word	Indian Language	Definition
Shadow	Chhaaya (Hindi) / Nizhal (Tamil) / Nerallu (Kannada)	A dark area formed when an object blocks light from a source.

Opaque	■■■■■■■■■ (Apardharshi)	Does not let any light pass through. Makes a dark, solid shadow.
Translucent	■■■■■■■■■■■■ (Ardh-pardharshi)	Lets some light through. Makes a faint, blurry shadow.
Transparent	■■■■■■■■ (Pardharshi)	Lets almost all light pass through. Makes little or no shadow.
Light Source	■■■■■■ ■■■■■ (Prakash Srot)	Anything that gives off light — Sun, torch, diya, tube light.
Gnomon	Shanku (Sanskrit)	The stick part of a sundial that casts a shadow to tell time. Ancient Indians used this!

■ SUCCESS CRITERIA FOR THIS SESSION

■ **Mastered**	Student correctly identifies all 3 material types (opaque/translucent/transparent) with Indian examples without prompting.
■ **Mastered**	Student predicts shadow size change (big/small) correctly for 4 out of 5 scenarios in the digital activity.
■ **Developing**	Student creates a shadow puppet and demonstrates understanding that opaque material gives the clearest shadow.
■ **Developing**	Student can explain in 1–2 sentences why the shadow changes size when the light moves.
■ **Exceeding**	Student connects shadow science to a cultural example (e.g., Tholpavakoothu or Jantar Mantar) independently.

■ STUDENT OBSERVATION JOURNAL — Print & Use

My Shadow Science Detective Journal

Material I tested	What I predicted	What I observed	Was I right?
Steel tiffin box			
Butter paper			
Glass of water			
My own object: _____			

Draw your shadow experiment below:

My biggest learning today:

____